ECE 310 Fall 2026
Lecture 12: Convolution as Template Matching

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Learning Objectives

- Explain how we may use convolution and matched filters to identify patterns in signals.
- Understand how we perform two-dimensional convolution.



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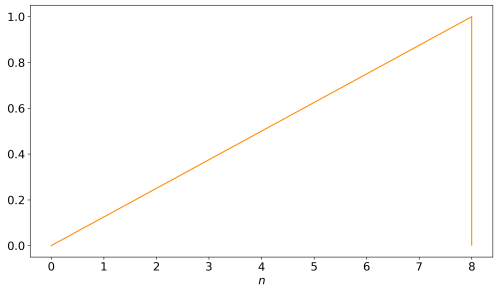
- Explain how we may use convolution and matched filters to identify patterns in signals.
- Understand how we perform two-dimensional convolution.

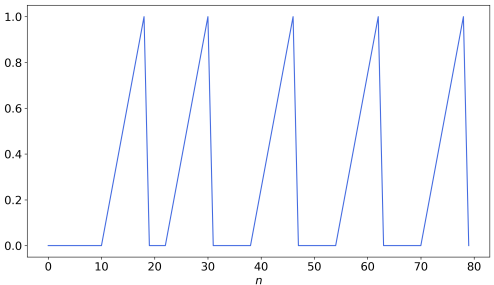
***Note*: this lecture will not be tested on homeworks or exams. This lecture is meant to be a break before we begin Fourier analysis to show an important time-domain application of LTI system models.**



Identifying Patterns in Signals

Suppose we have an example pattern $h[n]$ and a signal $x[n]$ that contains some number of shifted copies of this pattern. For example, the below sawtooth pattern $h[n]$ (left) and input signal $x[n]$ (right). How can we count and locate copies of $h[n]$?





Matched Filters

We can treat $h[n]$ as a *template* and see where it best lines up throughout $x[n]$. This can be accomplished using convolution!

Let $\tilde{h}[n] = h[-n]$. The resulting output is then

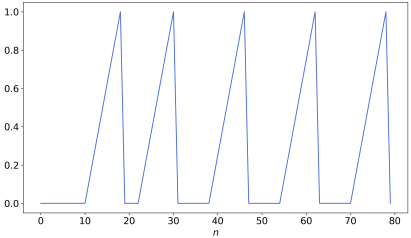
$$\begin{aligned} y[n] &= x[n] * \tilde{h}[n] \\ &= \sum_{k=-\infty}^{\infty} x[k]\tilde{h}[n-k] \\ &= \sum_{k=-\infty}^{\infty} x[k]h[k-n]. \end{aligned}$$

Whenever the template lines up perfectly with a repetition of the pattern, $y[n]$ will have its maximum value! We refer to $\tilde{h}[n]$ as the *matched filter* for this pattern.

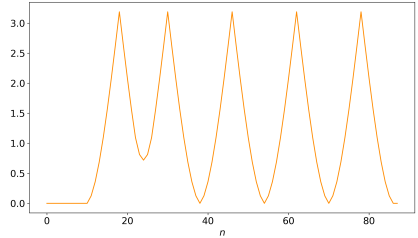


Matched Filters

Let's visualize this again:

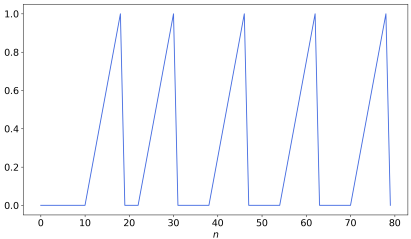


$$* \quad \tilde{h}[n] \quad =$$

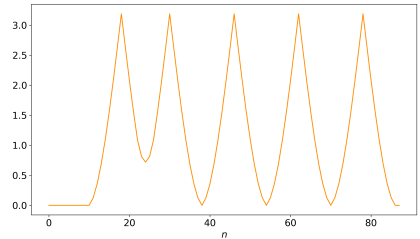


Matched Filters

Let's visualize this again:



$*$ $\tilde{h}[n]$ $=$



At each shift location, what mathematical operation does convolution perform?

Convolution in 2D

$$y[i,j] = x[i,j] * h[i,j] = \sum_{k=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} x[k,m]h[i-k, j-m]$$

0	-1	0
-1	2	-1
0	-1	0

$h[i,j]$

1	6	3	9	6	2	4
9	6	4	4	3	7	6
2	3	4	1	2	3	8
7	4	0	2	2	4	3
5	6	5	8	0	0	3
2	1	4	1	5	8	7
7	3	8	0	3	1	9

$x[i,j]$

$= (-1)(6) + (-1)(9) + (2)(6) + (-1)(4) + (-1)(3)$

$= -10$

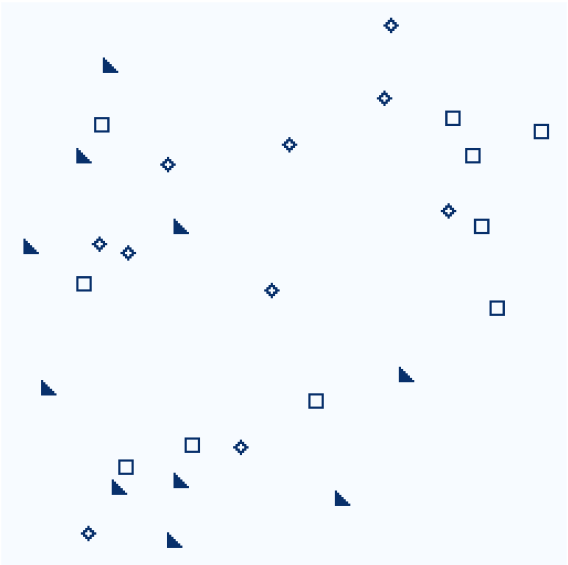
$= (-1)(2) + (-1)(5) + (2)(8) + (-1)(0) + (-1)(1)$

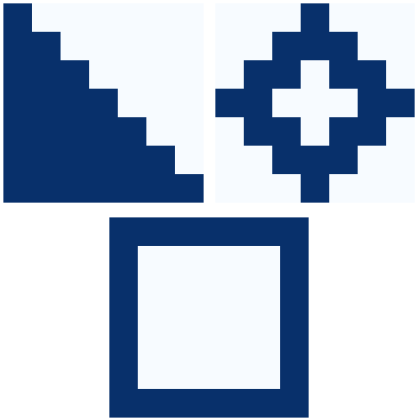
$= 8$

$= (-1)(6) + (-1)(3) + (2)(8) + (-1)(3)$

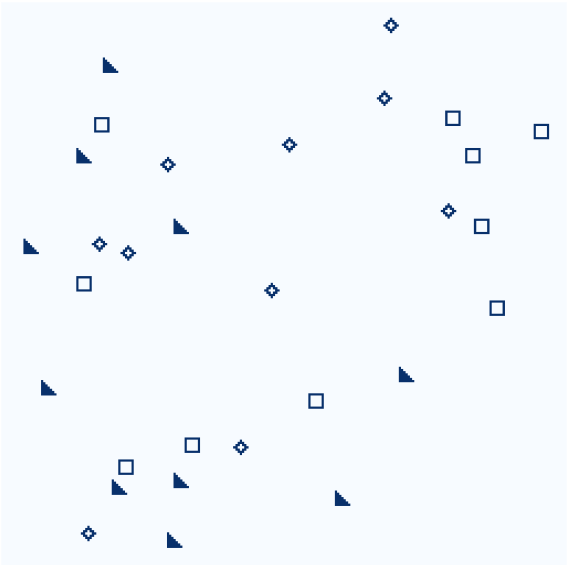
$= 4$

Template Matching in 2D

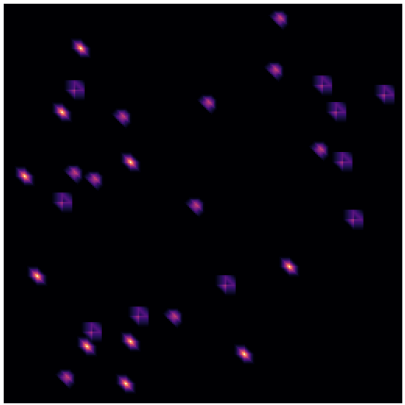




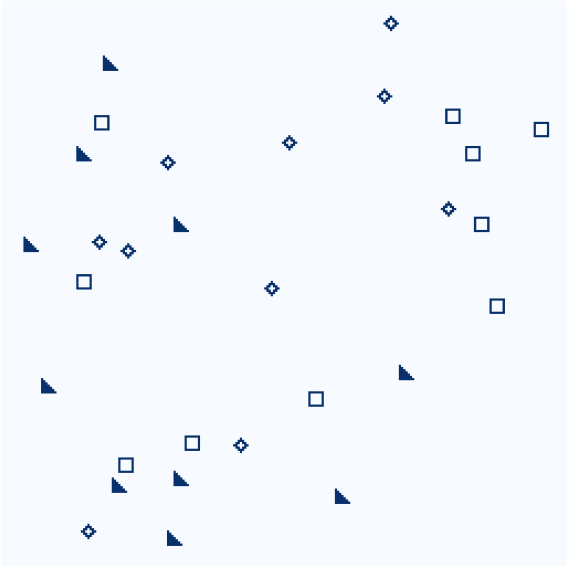
Template Matching in 2D



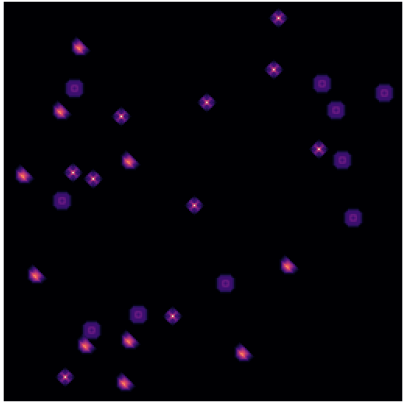




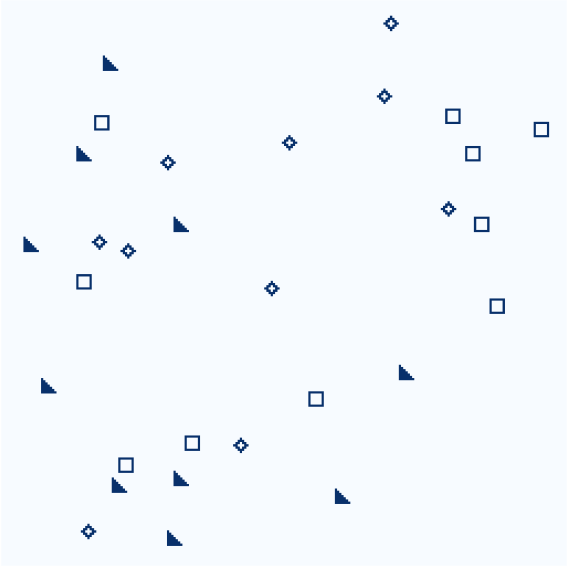
Template Matching in 2D



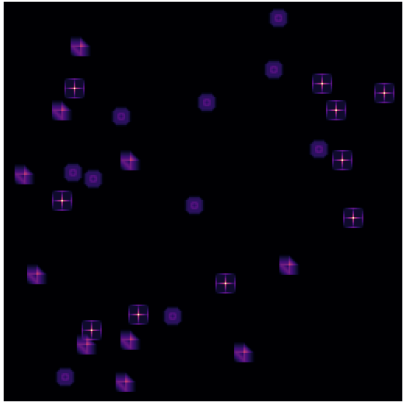




Template Matching in 2D







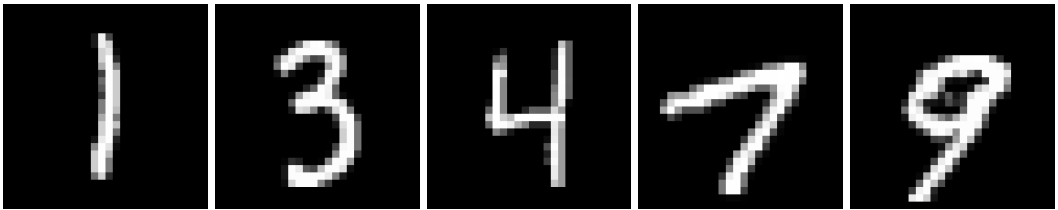
Templates as Features

We can extend this intuition to detecting small or basic features in more complicated structures like handwritten digits. Consider the following 2D filters or *kernels*.

$$h_v = \begin{bmatrix} -2 & 1 & 1 & 1 & -2 \\ -2 & 1 & 1 & 1 & -2 \\ -2 & 1 & \underline{\mathbf{1}} & 1 & -2 \\ -2 & 1 & 1 & 1 & -2 \\ -2 & 1 & 1 & 1 & -2 \end{bmatrix} \qquad h_h = \begin{bmatrix} -2 & -2 & -2 & -2 & -2 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & \underline{\mathbf{1}} & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ -2 & -2 & -2 & -2 & -2 \end{bmatrix}$$



Template as Features: Example Images



(a) Example 1. (b) Example 3. (c) Example 4. (d) Example 7. (e) Example 9.

Figure 1: Example images from MNIST handwritten digit dataset.

Template as Features: Vertical Filter Response

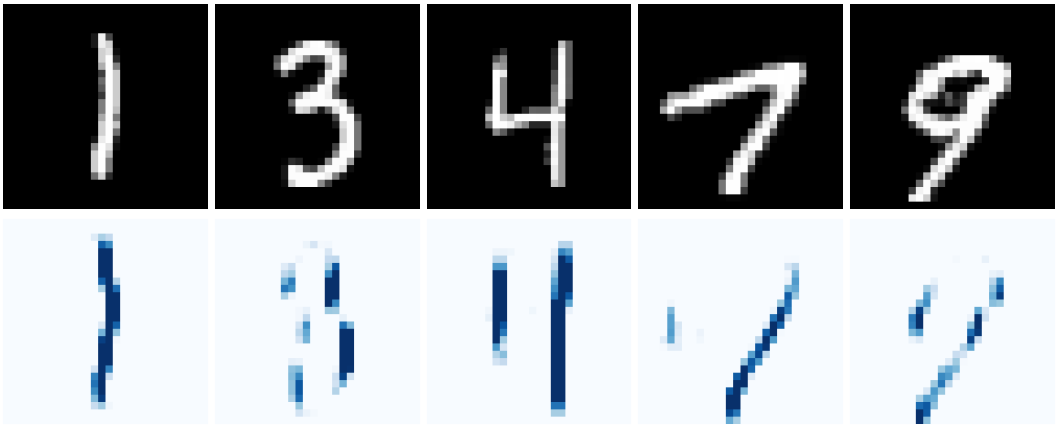


Figure 2: Response of example images to vertical kernel. Darker blue pixels have a stronger (higher value) response.

Template as Features: Horizontal Filter Response



(a) Example 1. (b) Example 3. (c) Example 4. (d) Example 7. (e) Example 9.

Figure 3: Response of example images to horizontal kernel. Darker blue pixels have a stronger (higher value) response.